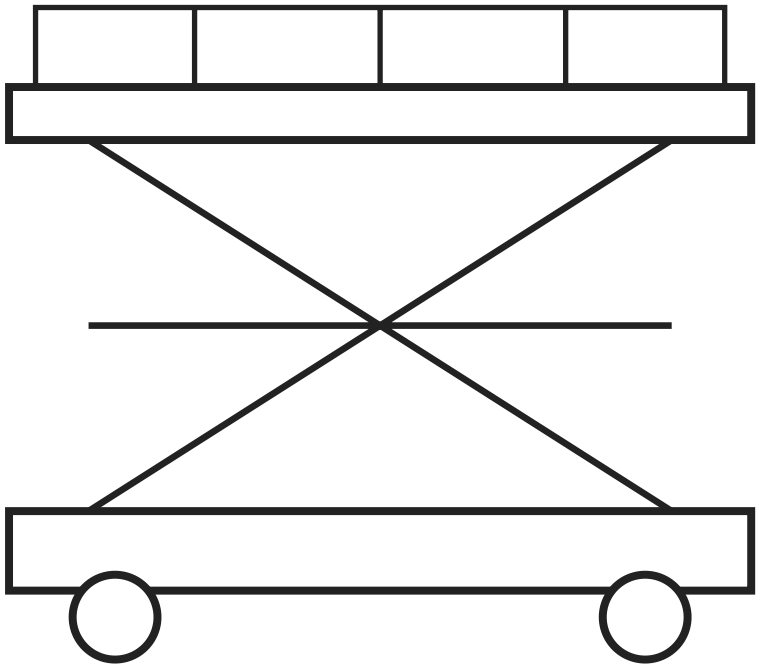
Skyjack

SERVICE MANUAL

Model SJIII 3219 Electric Scissor Lift

Rev. A, June 2024



Fleet units	FL-094, FL-095
Fleet	Mid-County Rental
Pages in excerpt	6

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1 Safety notes

WARNING

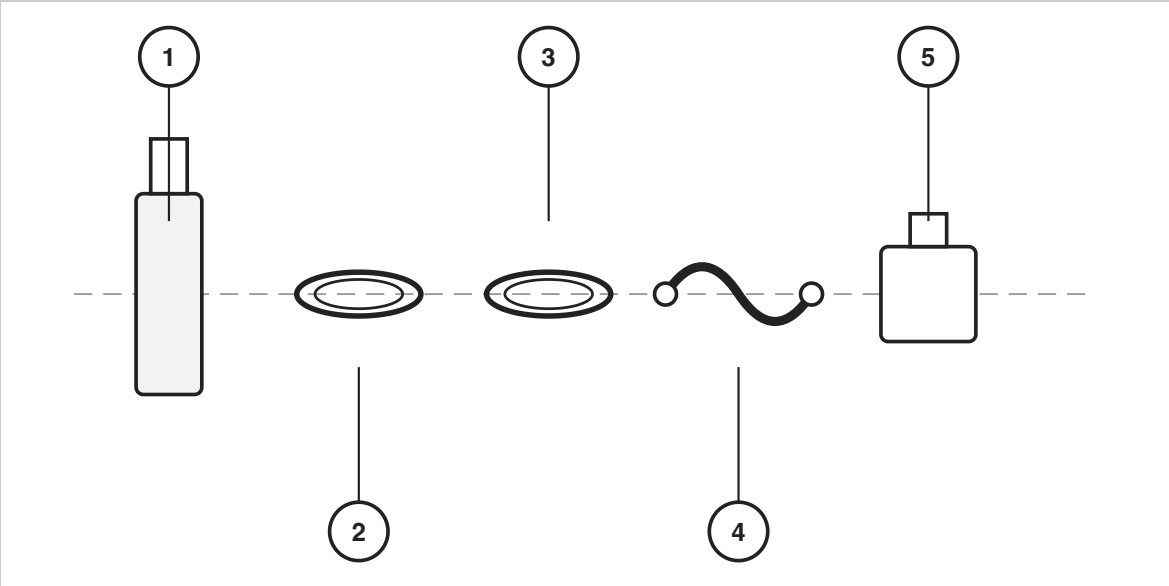
- Lower the platform fully and fit the maintenance prop before working under it.
- Turn the key switch off and disconnect the battery before any hydraulic work.
- Hydraulic fluid under pressure can penetrate skin. Relieve pressure before opening any line.
- Wear eye protection and gloves. Dispose of used oil and filters per local rules.

3.2 Lift cylinder seal replacement

Use seal kit SKJ-103100 and O-ring kit SKJ-103460.

1. Raise the platform, fit the maintenance prop and lower the platform onto the prop.
2. Turn the key switch off, disconnect the battery and tag the machine out of service.
3. Relieve system pressure by operating the manual lowering knob for 10 seconds.
4. Disconnect the hose at the cylinder base and cap both ends. Catch the fluid in a drain pan.
5. Remove the rod end and base pins and lift the cylinder clear with a hoist rated for 150 lb.
6. Unscrew the gland nut, withdraw the rod and remove the old seals. Fit the new SKJ-103100 seals, lubricated with AW32 oil.
7. Reassemble, torque the gland nut to 220 ft lb, refit the cylinder and bleed the system.
8. Cycle the platform 3 times to full height. Check for leaks and drift of more than 2 in over 10 minutes.

Figure 3-2 Lift cylinder, exploded view



ITEM	PART NUMBER	DESCRIPTION	QTY
1	SKJ-104210	Lift cylinder barrel and rod	1
2	SKJ-103100	Lift cylinder seal kit	1
3	SKJ-103460	Cylinder O-ring kit	1
4	SKJ-103455	Lift cylinder hose assembly	1
5	SKJ-103322	Lowering solenoid valve	1

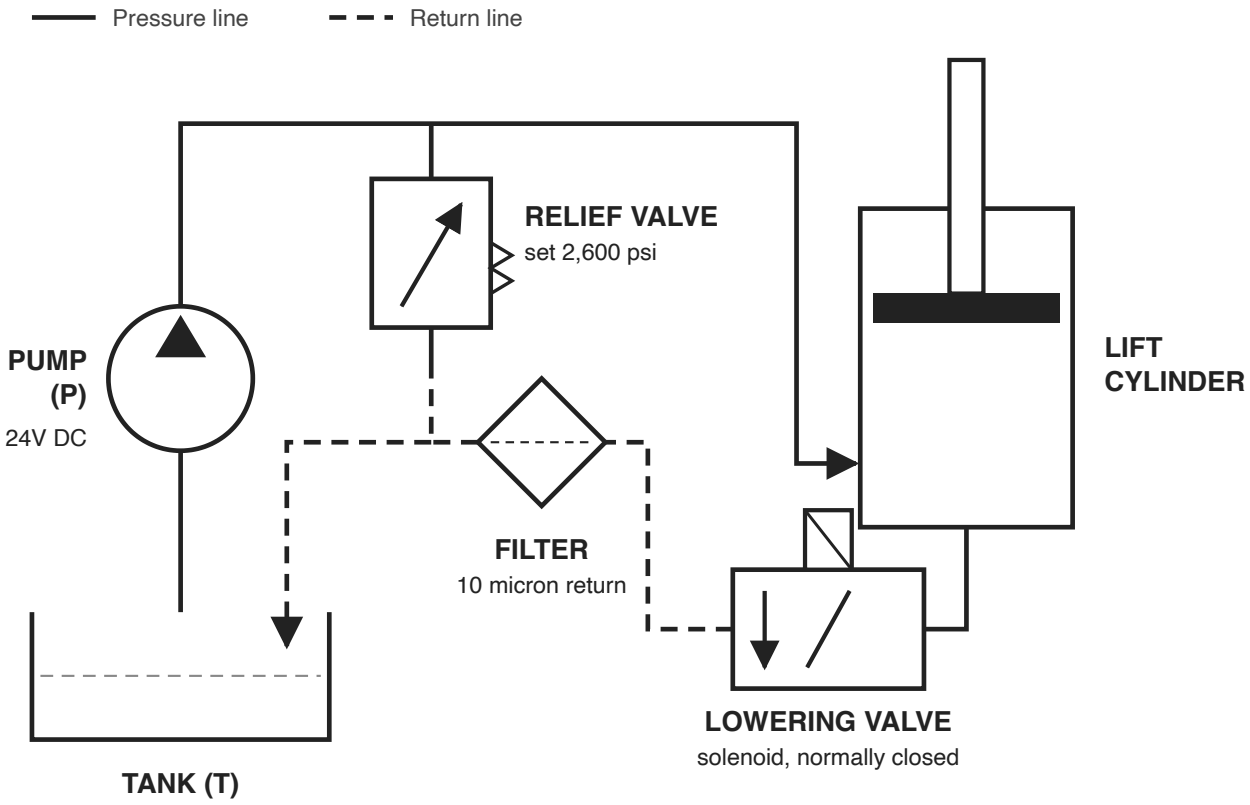
4 Specifications

ITEM	VALUE
Working height	25 ft (7.6 m)
Platform height	19 ft (5.8 m)
Platform capacity	550 lb (250 kg)
Machine weight	2,860 lb (1,297 kg)
Drive power	24V DC, 4 batteries 6V 220 Ah
Hydraulic reservoir	1.5 gal (5.7 L)
System relief pressure	2,500 psi (172 bar)
Lift time, up / down	26 s / 28 s
Hydraulic fluid	AW32 hydraulic oil

Service intervals

TASK	INTERVAL
Hydraulic filter element SKJ-105610	Every 250 h
Hydraulic oil change, AW32	Every 1,000 h or 2 years
Relief valve check	Every 500 h

5 Hydraulic schematic



COMPONENT	PART NUMBER
Pressure relief valve	SKJ-103278
Lowering solenoid valve	SKJ-103322
Filter element	SKJ-105610
Lift cylinder	SKJ-104210

6 Troubleshooting

SYMPTOM	LIKELY CAUSE	ACTION
Platform does not rise	Low battery charge	Charge batteries fully, check voltage at 24V
Platform does not rise	Relief valve set too low	Check pressure at test port, reset to spec
Platform drifts down	Lowering valve not seating	Clean or replace lowering valve, part SKJ-103322
Platform drifts down	Worn lift cylinder seals	Replace seals, see section 3.2
Slow lift speed	Clogged return filter	Replace filter element
Pump noisy	Low fluid level or air in system	Top up AW32 oil, cycle platform 3 times
Fluid on cylinder rod	Rod seal leak	Replace seal kit